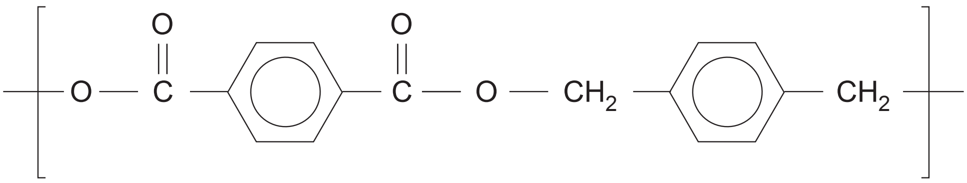
Examiner only

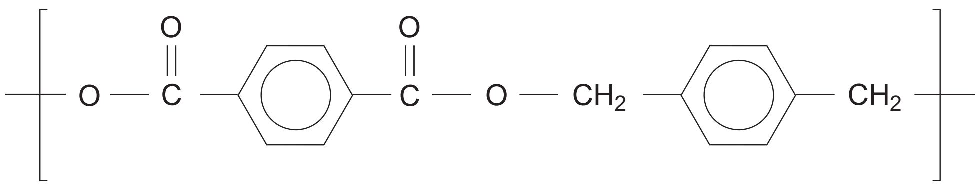
12. *(a)* A student was given a sample of a polyester **P**. He was told that the repeating unit was as follows.



(i) Draw a **circle** around an ester linkage in the repeating unit. [1]

(ii) Polyesters such as compound **P** can be hydrolysed by heating with aqueous sodium hydroxide.

Complete the balanced equation for this hydrolysis. [2]



+ .... NaOH

$\downarrow$

(iii) The formation of polyester **P** is an example of condensation polymerisation.

State how this type of polymerisation differs from addition polymerisation. Refer to both types in your answer. [1]

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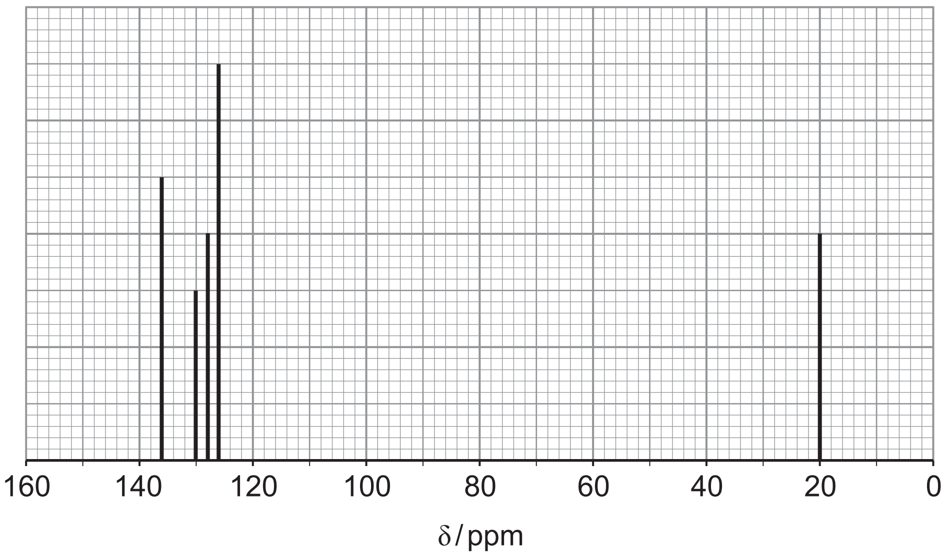
(iv) One of the raw materials used to make polyester **P** is methylbenzene. Under suitable conditions methylbenzene can be converted to a mixture of dimethylbenzene isomers and benzene.

$$2C_6H_5CH_3 \longrightarrow C_6H_4(CH_3)_2 + C_6H_6$$

Gas chromatography can be used to separate the dimethylbenzene isomers. One of the dimethylbenzene isomers has the ^{13}C NMR spectrum below.



Explain how this spectrum shows that the isomer is 1,3-dimethylbenzene. [3]

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(v) The isomer of dimethylbenzene used to produce polyester **P** is 1,4-dimethylbenzene. In industry this isomer is oxidised by air to produce benzene-1,4-dicarboxylic acid.

$$HOOC—C_6H_4—COOH$$

During this reaction a number of other aromatic oxidation products are made. One of these has the empirical formula C_4H_3O.

Suggest a structure for this product giving your reasoning. State how it could be produced in the reaction between 1,4-dimethylbenzene and atmospheric oxygen. [5]

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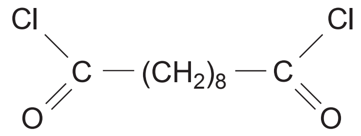
(b) (i) A laboratory experiment to demonstrate the formation of a polyamide uses decanedioyl dichloride.



State the name of a reagent used to produce this compound from the corresponding dicarboxylic acid, decanedioic acid. [1]

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(ii) After use, a bottle containing decanedioyl dichloride (M_r 239.1) can become contaminated with decanedioic acid.

Analysis of a 3.50 g sample of contaminated decanedioyl dichloride showed that it contained 0.977 g of chlorine by mass.

Calculate the percentage purity of this sample of decanedioyl dichloride. [2]

Percentage purity = ... %

(iii) Suggest how the decanedioyl dichloride had become contaminated with decanedioic acid. [1]

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END OF PAPER